

TAYYABA RAFIQ

AI Developer | Generative AI | Full Stack Developer

Karachi, Pakistan | 0342-2947573 | tayyabaaliowais@gmail.com | [GitHub](#) | [LinkedIn](#)

PROFESSIONAL SUMMARY

AI Developer and Full Stack Engineer with hands-on experience building AI agents, Retrieval-Augmented Generation (RAG) systems, and production-grade web applications. Skilled in designing AI-native documentation platforms, automation workflows, and full stack products using Next.js, FastAPI, and modern vector/database tooling. Combines strong prompt engineering and AI literacy expertise with practical, project-driven full stack development to build and ship real-world AI applications.

TECHNICAL SKILLS

AI & Generative AI: AI Agents • Prompt Engineering • Retrieval-Augmented Generation (RAG) • MCP • OpenAI • Gemini • Cohere • AI Automation • LLM Applications

Programming: Python • TypeScript • JavaScript

Frontend: Next.js • React • Tailwind CSS • HTML5 • CSS3

Backend: FastAPI • REST APIs • SQLAlchemy • Alembic

Databases: PostgreSQL • Neon • Qdrant • Sanity CMS

DevOps & Tools: Docker • Docker Hub • Git • GitHub • Vercel • Netlify

FEATURED AI PROJECTS

AI Documentation Platform & RAG Chatbot

- Developed an AI-native documentation platform for Physical AI & Humanoid Robotics.
- Built documentation site using Docusaurus.
- Implemented FastAPI backend with Retrieval-Augmented Generation (RAG).
- Integrated Cohere embeddings and Qdrant vector database.
- Used Neon PostgreSQL for persistent storage.
- Applied Spec-Driven Development for structured documentation.

Tech: Docusaurus • FastAPI • Cohere • Qdrant • Neon • PostgreSQL • RAG

Personal AI Employee

Designed an AI assistant capable of task automation and workflow orchestration.

- AI Agents
- MCP
- Gmail Integration
- File Watchers
- Approval Workflows
- AI Automation

Interactive AI Literacy Game — "Don't Be a Yes-Bot!"

Designed and developed an educational browser game that teaches AI sycophancy to teenagers through interactive, slider-based gameplay; built for the Governor House AI Summer Camp Faculty Challenge and received an A grade.

- Prompt Engineering
- Claude Collaboration
- Educational Game Design
- AI Safety
- Product Thinking
- Iterative Development

TodoFlow AI

AI-powered task management application.

Tech: Next.js • PostgreSQL • FastAPI

ADDITIONAL PROJECTS

- Rental Car Marketplace — Next.js, Tailwind CSS, Sanity CMS, deployed on Vercel
- EchoMind
- Portfolio Website
- Python Applications
- Docker Project

DEVOPS EXPERIENCE

Built and deployed a multi-service containerized application using Docker.

Tech: Docker • Docker Hub • FastAPI • Next.js • Multi-Service Architecture • Production Containers

EDUCATION

Governor Sindh Initiative for AI, Web3 & Metaverse (GIAIC) | Artificial Intelligence Program — Ongoing

Relevant areas: Generative AI, AI Agents, Prompt Engineering, Full Stack Development

M.Sc. Physiology | University of Karachi

CURRENT LEARNING

- OpenClaw
- Advanced AI Agent Systems
- Agent Workflows
- AI Engineering Best Practices

FUTURE ROADMAP

- Loop Engineering
- Advanced Multi-Agent Systems
- AI Evaluation
- Production AI Systems
- Enterprise AI Architecture